



1.6 Eigenvalues and eigenvectors (HL)

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For a square matrix A , a non-zero vector v is an eigenvector with eigenvalue λ if $Av = \lambda v$. Eigenvalues are found by solving the characteristic equation $\det(A - \lambda I) = 0$; for each eigenvalue, the corresponding eigenvector(s) solve $(A - \lambda I)v = 0$. Eigenvalues/eigenvectors reveal long-term behaviour of repeated matrix transformations (e.g. A^n), such as the dominant growth rate and stable direction of a system modelled by recurrence relations.

Worked example: follow the reasoning

Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

Solution: $\det\left(\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}\right) = 0 \rightarrow (4-\lambda)(3-\lambda) - 2 = 0 \rightarrow \lambda^2 - 7\lambda + 10 = 0 \rightarrow (\lambda-5)(\lambda-2) = 0$, so $\lambda = 5$ or $\lambda = 2$. For $\lambda = 5$: $-x + y = 0 \rightarrow y = x$, eigenvector $(1, 1)$. For $\lambda = 2$: $2x + y = 0 \rightarrow y = -2x$, eigenvector $(1, -2)$.

Quick check · multiple choice

For matrix A , an eigenvector v with eigenvalue λ satisfies which equation?

- A $Av = \lambda$
- B $Av = \lambda v$
- C $Av = v + \lambda$
- D $A + v = \lambda v$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A population model uses the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ to project two competing species' populations from one year to the next, given as a vector (x, y) . (a) Find the eigenvalues of A . (b) Find an eigenvector corresponding to the larger eigenvalue. (c) Explain what the larger eigenvalue represents for the long-term behaviour of the populations. [7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $\det\left(\begin{bmatrix} 3-\lambda & 2 \\ 1 & 2-\lambda \end{bmatrix}\right)=0 \rightarrow (3-\lambda)(2-\lambda)-2=0 \rightarrow \lambda^2-5\lambda+4=0 \rightarrow (\lambda-4)(\lambda-1)=0$, so $\lambda=4$ or $\lambda=1$. (b) For $\lambda=4$: $-x+2y=0 \rightarrow x=2y$, eigenvector $(2,1)$. (c) The dominant eigenvalue (4) is the long-term growth factor of the total population — as time increases the population vector aligns with direction $(2,1)$, and the total grows by a factor of approximately 4 each year since $|4|>1$.

Quick check answer

B. This is the defining relation for an eigenvalue-eigenvector pair.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.